Answers on
Booklet - 14

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/ = alternative and acceptable answers for the same marking point ; = separates marking points NOT = answers which are not worthy of credit ( ) = words which are not essential to gain credit ____ = (underlining) key words which **must** be used to gain credit ecf = error carried forward

1.

(a)(i)	$R = 50\ \Omega$	C1
	$I = 3.0 / 50$	C1
	current = 0.06 (A)	A1
(ii)	$P = VI / V^2 / R / I^2 R$	C1
	power = 3.0 x 0.06	
	power = 0.18 (W) (Possible ECF)	A1
(b)(i)	'Constant' temperature implied (wtte)	B1
	(Do not allow reference to Ohm's law or to 'heating')	
(ii)	1. 40 (Ω)	B1
	2. $A = \pi \times (1.0 \times 10^{-5})^2 = 3.1(4) \times 10^{-10}$ (m^2)	B1
	$R = \rho L/A$ (Allow other variant)	C1
	$40 = 5.4 \times 10^{-8} \times L / 3.1\ (4) \times 10^{-10}$ (Possible ECF)	C1
	$L = 0.23(3)$ (m) ≈ 0.23 (m)	A1
	Length is 9too) long, therefore must be coiled (wtte)	B1

2.

(a)	*interfere destructively:* the resultant wave has **reduced** (or zero) **amplitude** OR "waves cancel each other" OR shown on a diagram	B1	
	conditions: waves must meet **180°** OR π OR **1/2** λ out of phase OR in ANTI-phase 'crest of one meets trough of other' OR shown on diagram	B1	
	waves must have **equal amplitude** (not 'coherent' or 'same frequency')	B1	[3]
(b)(i)	for maxima **path difference = 0** OR λ OR **n**λ	C1	
	series of maxima occur at pts. where path diff = 0, λ, 2λ, 3λ, 4λ (OR when n = 0, 1, 2 , 3 ... OR n is an integer)	A1	[2]
(ii)	$x = \lambda D/a$ (allow $\lambda = ax/D$ or any valid form)	B1	[1]
(iii)	***x* DECREASES**	C1	
	to **HALF** previous value	A1	[2]
	{Allow full ecf from incorrect formula - e.g "x ∝ a hence x doubles" gets 2 marks- but give full credit for expected answer even if the candidate has quoted an incorrect formula in (ii)!}		
(c)	wavetrains must be **coherent** (OR const phase diff/in phase OR same frequency)	C1	
	an explicit reference to the SLITS or the TRANSMITTERS is needed: e.g. two **transmitters would not** be coherent (WTTE) OR **two slits in front of a transmitter** create coherent sources	A1	[2]

3.

a		when waves meet/cross/interfere/superpose/superimpose/overla /collide/interaction/mix/add resultant <u>displacement</u> = sum of <u>displacements</u>	B1 B1	Do not allow amplitude
b	i	(path difference =) $S_2P - S_1P = ½ \lambda$	B1	
b	ii	point **B** labelled above **P** a distance equal to **OP** OR below **O** a distance **2OP**	B1	judged by eye
b	iii	(fringe separation) x = 3.6 **x 2** = **7.2** mm recall of $\lambda = ax/D$ $a = \lambda D/x = (6.4\times10^{-7} \times 1.6)/7.2 \times 10^{-3}$ = **1.4×10^{-4} m** (1.78)	B1 C1 A1	many will forget to double 3.6 and will get **2.8×10^{-4} m** this scores **2 marks**

4.

(a) resistance = p.d./current (Allow voltage instead of p.d.) B2
(ratio of voltage to current scores 2/2)
(voltage per (unit) current scores 2/2)
(voltage per (unit) ampere scores 1/2)
($R = \frac{V}{I}$ scores 1/2 p.d. = current × resistance scores 1/2)

(b) $R = \frac{R_1 R_2}{R_1 + R_2}$ \ $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ C1
$R = \frac{10\times15}{10+15} = 6.0\ (\Omega)$ C1
resistance = 20 + 6 = 26 (Ω) (Possible ECF) A1

(c)(i) $P = VI$ \ I^2R \ $\frac{V^2}{R}$ C1
$P = 24 \times 0.028$ C1
power = 0.672 ≈ 0.67 (W) (-1 for 10^n error ∴ 670 (W) scores 2/3) A1

(c)(ii) (Thermistor's) temperature increases (due to electrical heating) [AW] B1
Resistance of <u>thermistor</u> decreases B1

5.

(i)	R = 1.28 / 0.80	C1
	resistance = 1.6(0) (Ω)	A1
(ii)	$R = R_1 + R_2$ / 1.60 = r + 1.10 / r = (1.28 – 0.8 x 1.1) / 0.8	C1
	r = 0.5(0) (Ω) ≈ 0.5 (Ω) (Possible ECF)	A1
(iii)	p.d = 1.10 x 0.80 = 0.88 (V) (Possible ECF)	B1

6.

(a)	$E = I(R + r)$ or $E = V + Ir$ / $R_T = \frac{1.5}{0.60} = (2.5)$ / $V_R = 1.8 \times 0.6$	C1
	1.5 = 0.60 (r + 1.8) / r = 2.5 – 1.8 / $r = \frac{1.5 - 1.08}{0.6}$	C1
	r = 0.70 (Ω) (Allow 1sf answer)	A1
(b)(i)	$P = \frac{V^2}{R}$ / $P = VI$ <u>and</u> $V = IR$	C1
	$36 = \frac{12^2}{R}$ / I = 3.0 (A) hence $R = \frac{12}{3.0}$	
	resistance = 4.0 (Ω) (Allow 1 sf answer)	A1
(b)(ii)	R_{series} = 30 (Ω)	C1
	$R = \frac{30 \times 4.0}{30 + 4.0}$ / $\frac{1}{R} = \frac{1}{30} + \frac{1}{4}$ / $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	C1
	resistance = 3.53 ≈ 3.5 (Ω) (Possible ecf)	A1
(b)(iii)	$I_{lamp} = \frac{36}{12}$ or 3.0 (A) / $I_{20\Omega} = \frac{12}{30}$ or 0.40 (A)	C1
	ratio = 7.5 / ratio = $\frac{30}{4}$	A1

7.

(a) similarity: any valid point e.g. (both have) vibrations, frequency, amplitude, wavelength, period, displacement (not velocity) B1

difference:

e.g. no energy transfer for standing waves }
neighbouring points vibrate in phase for standing waves } B1
only standing waves have nodes and antinodes } [2]
{allow standing waves are "trapped"/fixed/confined/don't move forward}

(b) (i) arrows show vertical oscillations B1
maximum amplitude at top {allow ecf for horiz.} B1
less in middle AND very small (or zero) at base B1 [3]
{allow 1 mark only for unlabelled diagram showing representation of amplitude}
{2 marks for unlabelled diagram plus an arrow}
{allow single headed arrows}

(ii) wavelength = 4 x 0.36 = 1.44m B1 [1]

(iii) recall of $v = f\lambda$ B1
$f = v/\lambda$ = 330/1.44 (allow ecf) = 229 (or 230) Hz B1 [2]

(iv) if open at both ends each end must be an antinode OR diagram B1
hence wavelength = 0.72m {allow ecf} C1
and frequency = 458 (or 460) Hz {allow ecf} A1 [3]

8.

a		pluck/stroke/disturb/vibrate/twang string	B1	Allow this mark for oscillating P or Q
b	i	correct diagram of fundamental i.e one loop	B1	Either full envelope or instantaneous pattern scores mark. Ignore any labelling of nodes and antinodes
	ii	wavelength = 2x1.2 = **2.4** m	B1	Allow ecf for candidates diagram
c		Correct standing wave drawn i.e 3 'loops'	B1	Either full envelope or instantaneous pattern scores mark
		All correct nodes labelled N for candidates diagram	B1	Allow lack of Ns labelled at P and Q if the others are correct. If all A and N are exactly reversed then award one of these two marks.
		All correct antinodes labelled A for candidates diagram	B1	

9.

(a)	the coil moves at **RIGHT ANGLES** OR across pulse direction	B1	
	and **OSCILLATES** (or WTTE)	B1	[2]
	{Allow 2 marks for fully labelled diagrams }		
(b)(i)	(air particles vibrate) **vertically** OR **up & down** OR **along tube**	B1	[1]
(ii)	**N** correctly labelled at the **base AND A** correctly labelled at the **top**	B1	[1]
(iii)	$0.4 = ¼\ \lambda \Rightarrow \lambda =$ **1.6 m**	B1	
	recall of $v=f\ \lambda$	C1	
	f = 320/1.6 = **200 Hz**	A1	[3]
	{allow ecf for candidate's λ - expect to see 400 Hz or 800 Hz each of which scores 2 marks }		
(iv)	any other **valid** mode of oscillation: **N at bottom AND A at top**	B1	
	all other **Ns AND As correctly labelled {allow ecf for cand's diagram}**	B1	[2]

10.

(a)	Correct circuit for both lamps in parallel (ignore ammeter here)	B1
	Ammeter placed correctly in <u>series</u> with **P**	B1
(b)(i)	The resistance of LDR/circuit changes (as light intensity changes)	B1
	When blade blocks light, resistance of LDR/circuit is large(r) (ora)	B1
	Correct statement about p.d (Possible ecf)	B1
(b)(ii)1.	(V = 5.0 – 3.0) 2.0 (V) (Allow 1 sf answer)	B1
(b)(ii)2.	$V = \frac{R_2}{R_1 + R_2} \times V_0$ | I = 2.0/2200 / 9.1× 10^{-4} (A)	C1
	(3.0 = $\frac{R}{R+2200} \times 5.0$) | ($R$ = 3.0 / 9.1× 10^{-4})	
	R = 3300 (Ω) | R = 3300 (Ω) Possible ecf	A1
	(For V_{LDR} = 2.0 V, R = 1.47 kΩ. This scores 1/2) (If 3.5 V given in **(b)(ii)1.**, then R = 940 Ω. This scores 2/2)	

11.

(a)(i)	Decreases	B1
(ii)	Decreases	B1
(b)(i)	$V = V_0 \times R_2 / R_1 + R_2$ $V = 5.0 \times 420 / (1000 + 420)$ p.d. = 1.4(*) (V) ≈ 1.5 (V) (Answer of 3.5(2) (V) scores 2/3) | $V = IR$ and $R = R_1 + R_2$ | $I = 5.0 / 1420 = 3.52 \times 10^{-3}$ (A) | p.d. = $3.52 \times 10^{-3} \times 420 \approx 1.5$ (V)	C1 C1 A1
(ii)	1. $R = R_1R_2 / R_1 + R_2 = 420 \times 50 / 420 + 50$ (=44.7 (Ω))	B1
	2. $V = 5.0 \times 45 / (1000 + 45)$ p.d. = 0.21(5) (V) ≈ 0.22 (V) (Allow V ≈ 0.2 (V) as long as working is shown) | $I = 5.0 / 1045 = 4.78(5) \times 10^{-3}$ (A) | p.d. = $4.78 \times 10^{-3} \times 420 \approx 0.22$ (V)	C1 A1

12.

(a)	$(\rho = \frac{RA}{L})$ $(\therefore \rho \rightarrow)$ $\Omega \times m^2/m$	M1
	(Cancellation of m leading to) $\rho \rightarrow [\Omega\ m]$	A0
(b)(i)	$\rho = \frac{RA}{L}$ (Any subject for equation) (Possible credit from (a))	C1
	$A = \pi r^2 = \pi \times (0.62 \times 10^{-3})^2$ $(= 1.208 \times 10^{-6}\ (m^2))$	C1
	$R = \frac{6.8 \times 10^{-6} \times 0.32}{\pi \times (0.62 \times 10^{-3})^2}$	C1
	$R = 1.8\ (\Omega)$	A0
(b)(ii)	$V = \frac{R_2}{R_1 + R_2} \times V_0$ $V = \frac{1.8}{2.8} \times 5.0$ $V \approx 3.2$ (V) (1.8 (V) scores 2/3) | $V = IR$ and $R_T = R_1 + R_2$ | $I = 5.0 / 2.8 = 1.786 \approx 1.8$ (A) | $V = 1.786 \times 1.8 \approx 3.2$ (V)	C1 C1 A1
	(Allow ECF from (b)(i) but –1 mark for not using given value of 1.8 Ω)	
(b)(iii)	The p.d. increases	B1
	because the resistance (of wire) increases (as it gets thinner \ longer)	B1

13.

a		Any two from: • Packet of energy / quantum of energy / electromagnetic radiation • Travels at the speed of light / 3.0×10^{8} ms^{-1} • Has no charge • Energy given by: $E = hf$ / $E = hc/\lambda$ • Can travel through a vacuum / free space	B1×2	**Must show ticks on the script to indicate where marks are being awarded** **Allow** has zero mass **Do not allow** wave effects such as polarisation/diffraction/reflection.
b	i	$E = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{5.5\times10^{-7}}$ $E = 3.6(2)\times10^{-19}$ (J)	C1 A1	**Allow** The first mark for: $E = hf$ and $f = \frac{3.0\times10^{8}}{5.5\times10^{-7}} (= 5.46\times10^{14})$
	ii	total power = $6.3\times10^{7}\times6.1\times10^{18}$ (= 3.843×10^{26}) number per sec = $\frac{3.843\times10^{26}}{3.62\times10^{-19}}$ (Possible ecf) number per sec = $1.0(6)\times10^{45}$ (s^{-1})	C1 A1	**Allow** 1 mark if candidate gets as far as the second step **Allow** 2 marks for a bald answer of 1.1×10^{45} (s^{-1}) – 2sf or more **Special case** 1 mark for 1.74×10^{26} (s^{-1}) (i.e. per m^2)
	iii	Infrared / IR	B1	

14.

(a) (1 eV) is the work (done) / energy gained / transformed by an electron travelling / accelerated through / across a p.d. / voltage of 1 V **B1**
(Allow a proton instead of an electron)
1 eV = 1.6×10^{-19} (J) (Ignore omission of unit but NOT 1.6×10^{-19} C) **B1**

(b) energy, photon and (photo)electron respectively **B1** × 3
(Allow 'momentum' / 'charge' instead of 'energy' for the first marking point)

(c)(i) energy of photon = work function energy / $hf = \phi$ **B1**
$\phi = 6.63\times10^{-34}\times3.0\times10^{8} / 3.2\times10^{-7}$ / $\phi = 6.63\times10^{-34}\times9.375\times10^{14}$ **C1**
$\phi = 6.22\times10^{-19}$ (J) $\approx 6.2\times10^{-19}$ (J) **A1**

$\phi = 6.22\times10^{-19} / 1.6\times10^{-19}$
$\phi = 3.88$ (eV) ≈ 3.9 eV Possible ECF **A1**

(ii) No (photo)electrons **B1**
Photon energy is less than the work function (energy) / The frequency (of the radiation) is less than the threshold frequency / Photons heat the metal **B1**

15.

(a)	Quantum of energy \ radiation \ Packet of energy	B1
(b)(i)1.	$\Delta Q = I\Delta t$ \ $1.2 \times 10^{-7} \times 5$ (Any subject and no need for Δ notation)	C1
	charge = 6.0×10^{-7} (C) (Allow 6×10^{-7} (C))	A1
(b)(i)2.	number = $\frac{6.0\times10^{-7}}{1.6\times10^{-19}} = 3.75 \times 10^{12} \approx 3.8 \times 10^{12}$ (Possible ECF)	B1
(b)(ii)	$E = hf$ \ $E = 6.63 \times 10^{-34} \times 7.0 \times 10^{14} = 4.64 \times 10^{-19} \approx 4.6 \times 10^{-19}$ (J)	C1
	$hf = \phi + KE_{max}$ (Allow other subject for photoelectric equation)	C1
	$KE_{max} = (4.64 - 3.5) \times 10^{-19} = 1.14 \times 10^{-19} \approx 1.1 \times 10^{-19}$ (J)	A1
(b)(iii)1.	Energy of photon is the <u>same</u>	B1
(b)(iii)2.	KE of electron is the <u>same</u>	B1
(b)(iii)3.	The current <u>doubled</u> (as there are twice as many photons)	B1